Research Statement

Nahee Kim

Humans begin learning language from the moment they are born by observing the sounds and actions around them. Even though young children cannot explicitly explain the grammar of the language they use, they learn the relationships between recurring expressions and situations, allowing them to understand and produce appropriate sentences. Understanding language goes beyond knowing the meanings of individual words. The same expression may require entirely different actions depending on who says it, when it is said, and with what intention. Language is ultimately a means of guiding action, and acting appropriately requires the ability to understand the context in which an utterance is situated.

I experienced university life across the period before and after the emergence of ChatGPT, which allowed me to directly observe this distinction. When using early language models, the greatest difficulty was that they could generate grammatically natural sentences while misunderstanding what the user had actually requested and producing irrelevant answers. At the time, I regarded this merely as an inconvenience caused by insufficient model performance. However, as language models become connected to search engines, external documents, personal information, and tools, and evolve into agents capable of carrying out actions autonomously, misunderstanding context no longer remains a minor response error. If a model cannot distinguish between a user's instruction and a sentence contained in external data, or if it deviates from the original objective as an interaction progresses, it may perform unintended actions and create security risks.

These observations motivate the questions that guide my research: How do AI models internally distinguish instructions, evidence, and irrelevant or adversarial information while forming context over time? How are these relationships represented and transformed in hidden-state space? Can changes in internal trajectories reveal when a model begins to depart from the user's intention, and can those trajectories be redirected before the deviation results in an unsafe action?

1 Research Conducted During My Master's Program

1.1 TST-Mamba

NLOS Transient Data as a Physical Language and Mamba-Based Selective Signal Learning

When I began conducting research in a computer vision laboratory, I came to view NLOS transient data as another form of language that a model must interpret. In non-line-of-sight (NLOS) imaging, a camera cannot directly observe the target scene. Instead, the hidden scene must be inferred from the arrival times and intensities of light that has undergone multiple reflections between a visible wall and a concealed object. Each measurement is highly noisy and ambiguous in isolation, yet the relationships among signals distributed across time and space indirectly encode the structure of the hidden scene. Just as a word or sentence in natural language is interpreted within the context of a conversation, a transient signal acquires physical meaning only through its surrounding spatiotemporal relationships.

I did not regard this task as a simple denoising problem. The central challenge was to enable the model to distinguish signals required for reconstruction from irrelevant disturbances while preserving the fine spatiotemporal structures needed to infer the hidden scene. To address this challenge, I focused on Mamba, a state-space model capable of efficiently learning long-range sequential dependencies.

Mamba processes inputs sequentially, accumulates prior information in a hidden state, retains information that is relevant to the current input, and suppresses information that is unnecessary. I interpreted this mechanism not as the independent processing of individual measurements, but as a process of forming context in which the meaning of the current signal is determined in relation to preceding information. In NLOS transient data, a single time bin does not provide sufficient information about the hidden scene. Determining whether a particular measurement is a valid photon response returned from an object or merely background noise requires jointly considering temporal changes before and after that measurement and signals from neighboring spatial locations. I therefore used Mamba to scan the temporal sequence at each spatial location and treated the final hidden state, in which information from the full temporal interval had been selectively accumulated, as a compressed bottleneck representation. This representation is not a simple average or pooling result, but a learnable summary formed by retaining or suppressing different information depending on the input. Spatial Mamba then sequentially processes the compressed bottleneck tokens so that the model can learn, in latent space, how the signal at each location relates to signals at neighboring locations. In this architecture, Temporal Mamba forms context accumulated over time, whereas Spatial Mamba integrates context across spatial locations.

However, compressing context alone was not sufficient to faithfully reconstruct the original signal. The compressed representation provided the overall spatiotemporal structure but could not preserve every sharp temporal peak and weak photon return required for NLOS reconstruction. I therefore designed a structure that transfers the full temporal features through a skip path and combines them with the bottleneck code containing spatial context. Whereas the bottleneck path provides the global context needed to interpret the signal, the skip path preserves local and fine-grained evidence that may be lost during compression. By combining these two paths, the model can reconstruct a clean transient using both global relationships and detailed observational evidence, without relying solely on context or responding only to isolated signals. Through this analysis, I formalized TST-Mamba as a Mamba-based skip-connected denoising autoencoder. Temporal Mamba encodes a noisy transient sequence into a compressed representation, Spatial Mamba forms spatial context in latent space, and the skip–latent fusion decoder combines accumulated context with the original temporal details to reconstruct the signal. This work showed that Mamba can be interpreted not merely as an architecture for efficiently processing long sequences, but as a structure that learns what to remember, what to suppress, and which information to use for its final judgment.

Through this research, I came to believe that "context" does not simply mean providing a model with a large amount of information. Understanding context is a process of selecting the information required for the current task, forming relationships among information that is separated in time or space, suppressing irrelevant disturbances, and preserving the detailed evidence necessary for the final judgment. In NLOS transients, an incorrect selection may prevent the model from distinguishing signal from noise or, conversely, cause it to remove weak but valid signals. Similarly, in natural language models, misinterpreting the role or source of information may blur the distinction between user instructions and external data and cause the model's behavior to deviate from its intended task. This experience expanded my interest beyond the restoration of physical signals toward the analysis of how context is formed and transformed within AI models and how those changes influence model decisions and behavior.

1.2 RiemannMamba

A Geometric Interpretation of Mamba's Internal Representations

Through the TST-Mamba project, I became interested in how Mamba selectively retains and suppresses input information while forming task-relevant context in its hidden state. This led me to ask a further question: As Mamba processes an input sequence, which tokens does it treat as more important, and which tokens leave information in the internal state for longer periods so that it continues to influence subsequent decisions? In other words, even if all tokens are provided in the same order, might each token possess a different "duration" and degree of influence within the model?

The final output alone cannot reveal which tokens were quickly forgotten and which continued to affect later hidden-state changes. Existing explainable AI methods primarily estimate the importance of individual input features to the final output. In a state-space model such as Mamba, however, it is also necessary to understand when information is accepted, how long it is retained, and in what direction it changes as it is combined with other information. I therefore viewed this process not as a collection of independent points in a high-dimensional representation space, but as geometric trajectories that progress at different rates for different tokens.

This perspective was inspired by the concept of proper time in Einstein's general theory of relativity. In general relativity, time is not an absolute measure shared identically by every observer. The proper time experienced by an observer differs according to the geometry of spacetime shaped by mass and energy and the path followed by that observer. Encountering this concept led me to ask whether the tokens processed by Mamba might also possess an "internal time" distinct from their externally assigned order. All tokens pass through the same network depth and scan order, but the information contained in each token and its surrounding context differ. Some tokens may therefore induce only a small change in the hidden state and be processed with limited internal evolution, whereas ambiguous or complex tokens may produce larger changes in the model's internal representations and continue to influence subsequent processing. Thus, even when the external computational steps are identical, the magnitude of change and the accumulated path experienced by the model in representation space may differ across tokens. I interpreted this as an internal time that flows at a different rate for each token.

This idea connects naturally to Mamba's core variable, the input-dependent step size Δ. In Mamba, Δ adjusts the degree of state transition according to each input and determines how strongly the hidden state changes for each token. It has generally been interpreted as a computational gate that selectively passes or suppresses information. I instead considered whether Δ could serve as a geometric scale that determines how each change should be measured in the model's hidden-state space. To formalize this idea, I treated the hidden-state space of each token as a Riemannian manifold and constructed state-dependent metric weights from Δ. The hidden-state change before and after the selective scan is then measured not as a simple Euclidean distance, but as a local line element weighted by this metric. By accumulating the changes produced at each layer, I defined the length of the path traveled by the model in representation space while processing an input as its internal time τ. Thus, τ does not represent a token index or actual computation time; it is the accumulated amount of geometry-weighted hidden-state evolution induced by an input. From this perspective, I hypothesized that easy inputs would induce relatively small internal state changes and accumulate shorter internal times, whereas difficult or error-prone inputs would form more complex hidden-state trajectories and accumulate longer internal times. In practice, the internal time measured by RiemannMamba showed a strong relationship with input difficulty and was larger for misclassified and corrupted inputs. These findings demonstrate that even when all inputs pass through the same computational depth, the amount of change they induce within the model may differ.

This research also showed that assessing model reliability solely through final-output accuracy or confidence is insufficient and that the internal-state changes formed while processing an input can serve as an important reliability signal. RiemannMamba's internal time τ represents not the number of computational steps, but the accumulated geometric changes induced by each input in Mamba's hidden-state space. Inputs that were difficult to classify, misclassified, or affected by distribution shift accumulated longer internal times, allowing the likelihood of model error to be evaluated from a perspective distinct from final confidence. In addition, Tau-CAM, which visualizes the spatial distribution of internal time, made it possible to observe which input regions induced large changes in the model's internal state.

1.3 Driver Handover Readiness Estimation via Bio–Vehicle Signal Fusion and Safety–Oriented Re-decision

Safe Decision-Making in Human–Machine Collaboration

In conditional automated driving, transferring control of the vehicle back to the driver is not merely a driver-state classification problem, but a decision-making problem that must safely coordinate collaboration between a human and an automated system. Before handing control to the driver, the vehicle must determine whether the driver has recognized the current driving situation and is actually ready to resume driving. Transferring control too early to an unprepared driver increases the risk of an accident, while excessively delaying a necessary handover may also compromise safety.

I conducted a multimodal learning study that estimated driver handover readiness by jointly using the driver's physiological signals and vehicle signals. Physiological signals reflect the driver's cognitive and physical arousal and readiness, whereas vehicle signals describe the current dynamic state of the vehicle, including speed, steering, acceleration, and deceleration. Because the two modalities provide information about different aspects of the situation, neither alone is sufficient to determine whether a safe handover is possible. I therefore integrated the two signal types to evaluate the states of the driver and vehicle together and to achieve more stable readiness estimation than would be possible with a single modality. However, high classification accuracy alone cannot guarantee safety in real driving environments. The driver's condition and the vehicle's driving state continue to change, meaning that a handover decision that was appropriate at the initial decision point may later become unsafe. I therefore studied a safety-oriented re-decision structure that does not fix the initial readiness estimate as the final decision, but instead reassesses handover feasibility using newly observed physiological and vehicle signals. When the model's prediction is uncertain or the driver's readiness is judged insufficient, the system withholds or reevaluates the transfer of control, preventing a single erroneous prediction from immediately leading to a hazardous action. This research taught me that the role of AI in systems where humans and machines collaborate is not simply to replace the human, but to coordinate when and how actions should be taken while jointly considering the human's condition and the system's limitations. In safety-critical environments, the ability to recognize uncertainty and revise an earlier decision in response to a changed state is more important than trusting a single prediction. This experience led me to view trustworthy AI not simply as a high-accuracy model, but as a system that can collaborate with people while preserving human control and safety.

1.4 Integration of My Research Experience

TST-Mamba examined how a model selects and preserves information in spatiotemporal data, while Riemann-Mamba investigated how the selected information shapes hidden-state dynamics and how those dynamics relate to task difficulty and reliability. The Driver Handover project extended this perspective from internal representations to action, showing that a decision must be reevaluated when the state of the human or the environment changes. Together, these studies led me to view trustworthy AI as a process rather than a property of a single output: reliable behavior depends on how information is selected, how internal states evolve, and whether a system can revise its decision as the surrounding context changes.

Building on this integrated perspective, my future research will focus on identifying early signs of task or intention drift from hidden-state trajectories and on testing whether targeted interventions can redirect those trajectories before they lead to unreliable or unsafe behavior.

2 Future Research

Geometry-Aware White-Box Analysis and Control of AI Systems

The most fundamental question I want to investigate is: What actually happens inside an AI model as it forms, revises, or abandons an interpretation? My future research will examine this question through the geometry of evolving hidden representations. The proposed framework is not limited to a single architecture. In state-space models, hidden states form explicit sequential trajectories over time, while in Transformer-based models, representations can be examined as trajectories that evolve across layers, tokens, and interaction turns. I will use state-space models as a tractable setting for developing this framework, while examining its relevance to language models and agentic systems from the early stages of the research.

My goal is to develop white-box methods that reveal not only which input features influenced a final prediction, but also how instructions, evidence, and external information were internally distinguished and combined during decision formation. This distinction becomes a security problem when a model processes both trusted user instructions and untrusted external data. An adversary may embed an instruction within a document, message, webpage, or tool output so that the model gradually abandons the user's original objective while its final response remains apparently plausible. I aim to determine whether such manipulation produces detectable changes in hidden-state geometry and whether those changes can support intervention before the model performs an unsafe action.

2.1 Hidden-State Geometry as a Basis for White-Box AI

Deep learning models represent hidden states as vectors $h \in \mathbb{R}^d$, and their internal changes are often analyzed using Euclidean notions of distance and direction. However, representing a state with Euclidean coordinates does not require the meaningful geometry of the representation space to be globally flat. Language, images, time series, and physical signals contain nonlinear and structured relationships that may be better described through a curved latent manifold.

I therefore hypothesize that model computation involves two coupled processes: the evolution of the hidden state and the evolution of the geometry through which that state is interpreted,

$$h_{k+1} = F(h_k, x_k), \qquad g_k = G(h_k, x_k),$$

where g_k denotes a local, input-dependent metric and k may index time steps, layers, tokens, or interaction turns. Under this view, an input does not merely move a representation through a fixed space. It may also alter the local geometry that determines which directions are important, which states remain close, and which relationships continue to influence later computation.

This perspective is formally inspired by the relationship between matter and geometry in general relativity. I do not claim that neural networks literally obey physical laws. Rather, I draw on the principle that the conditions within a system may reshape the geometry through which its dynamics unfold. Analogously, information and accumulated context may reshape latent geometry, while hidden states evolve along the resulting paths.

This framework also suggests a geometric interpretation of memory. Long-range dependencies may rely not only on retaining earlier state values or providing later access to token representations, but also on forming a geometry in which contextually related states remain effectively close. States that are distant in sequence order or Euclidean coordinates may have a short task-dependent geodesic distance. From this perspective, useful representation learning involves learning not only useful coordinates, but also a useful geometry that organizes internal relationships and trajectories.

A geometry-weighted trajectory length can be expressed conceptually as

$$\tau = \sum_k \sqrt{\Delta h_k^\top g_k \Delta h_k}, \qquad \Delta h_k = h_{k+1} - h_k.$$

Here, τ is not physical time. It represents the accumulated amount of internal change measured under the geometry induced by the input and context. Together with curvature, changes in direction, and geodesic relationships, this quantity may provide an interpretable description of how a model forms, preserves, and revises its internal context.

Rather than examining only the magnitude of an activation change at a single point, I aim to study how internal changes accumulate across an entire trajectory, where divergence begins, and how the relationships among internal states are reorganized. This may make it possible to localize when and where external information begins to reshape the model's interpretation of its original task.

2.2 Research Questions under Adversarial Context Manipulation

I will organize this research around three experimentally falsifiable questions.

Q1. Does input-adaptive geometry reveal task drift more clearly than conventional internal-state measurements?

I hypothesize that clean task execution, benign context variation, and instruction injection produce distinguishable hidden-state trajectories. I will examine whether geometry-aware quantities such as trajectory length, curvature, changes in direction, and local metric variation provide information beyond Euclidean activation differences, internal probes, output confidence, and output-level safety checks. This hypothesis would not be supported if the geometric description provides no consistent advantage in detecting or localizing task drift across architectures and attack conditions.

Q2. Are geometric drift signals robust to an adaptive attacker?

A detector that succeeds only against fixed or previously observed injections would be insufficient as a security mechanism. I will therefore consider an attacker who is aware that internal trajectories are being monitored and attempts to alter model behavior while minimizing detectable geometric deviation. I will investigate whether the distinction between clean and manipulated trajectories remains stable when the attack is adapted to evade the detector. If an attacker can consistently conceal the internal deviation while preserving attack success, the proposed geometric signal should be regarded as insufficient for reliable defense.

Q3. Can geometry-guided intervention causally redirect an unsafe trajectory?

Detection alone does not establish that the identified internal structure plays a functional role in model behavior. I will therefore investigate whether targeted intervention on hidden representations can restore the original task, reduce the influence of injected instructions, or prevent an unsafe external action. This hypothesis would be supported only if the interventions produce predictable improvements in security while preserving performance under benign conditions. Otherwise, the identified geometry may be correlated with task drift without being causally involved in it.

These questions connect the theoretical idea of input-dependent geometry to a concrete security objective. The purpose is not simply to create a new visualization of latent space, but to determine whether internal geometry can distinguish legitimate context formation from adversarial manipulation, remain informative when an attacker attempts to conceal the manipulation, and support causal control of model behavior.

2.3 Safe Intervention and Human Oversight

Language models and agents may receive a trusted instruction from a user while simultaneously processing untrusted documents, emails, webpages, messages, or tool outputs. If a model internally reinterprets external content as a command, it may depart from the user's objective before this deviation becomes visible in its final response or external behavior.

I aim to use hidden-state geometry to identify when this reinterpretation begins. A persistent change in trajectory, the breakdown of relationships formed under the original instruction, or the emergence of a new instruction-like direction may indicate that external information is taking control of the model's task. Unlike a safety mechanism that evaluates only a completed response, this approach would examine the process through which the unsafe decision is being formed.

Detection should lead to an appropriate re-decision. In my Driver Handover research, I studied a system in which decision authority should return to the driver when the state of the human or the environment makes autonomous operation unreliable. I aim to extend this principle to AI agents. When an internal trajectory becomes uncertain or adversarially manipulated, the system may steer its representation toward the original task, withhold a tool call, request clarification, isolate untrusted content, or return the decision to a human.

My legal studies have also shaped this perspective. As AI systems gain access to personal information, external tools, and consequential decision-making authority, meaningful human oversight should not be treated as an interface added only after deployment. It should be supported by the model's ability to reveal when its own internal process has become unreliable and to preserve opportunities for human intervention before an irreversible action occurs.

2.4 Expected Contributions and Research Direction

This research will develop geometry-aware white-box methods across both state-space models and Transformer-based systems. State-space models will serve as a controlled setting in which sequential dynamics are explicitly represented, allowing the theoretical framework to be formulated and examined clearly. Language models and agents will not be deferred to a distant extension; they will provide an early and essential setting for examining instruction–data separation, prompt injection, task drift, and adaptive attacks.

I aim to make three main contributions. First, I will develop a theoretical framework that treats internal representations as trajectories evolving through input-dependent geometry. Second, I will construct interpretable signals that reveal when trusted instructions, external evidence, and adversarial information begin to alter the model's internal task representation. Third, I will develop and evaluate white-box control methods that redirect unsafe trajectories, block risky actions, or return decision authority to a human.

Ultimately, I aim to develop AI systems whose decision formation can be observed, experimentally tested, and controlled. Such systems should not merely explain a completed answer after the fact. They should reveal when and how their internal reasoning begins to depart from a trusted objective, remain robust when an attacker attempts to conceal that departure, and support intervention before the deviation results in harmful action.